

CS3002E Compiler Design

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Lecture #18

Bottom-Up Parsing - 7: LALR Parsing

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LR(1) Item: Review

Items of the form $[A \rightarrow \alpha.\beta, a]$

$[A \rightarrow \alpha., a]$ - reduce by $A \rightarrow \alpha$ when lookahead is a

$[A \rightarrow \alpha. b\beta, a]$ - shift input symbol b

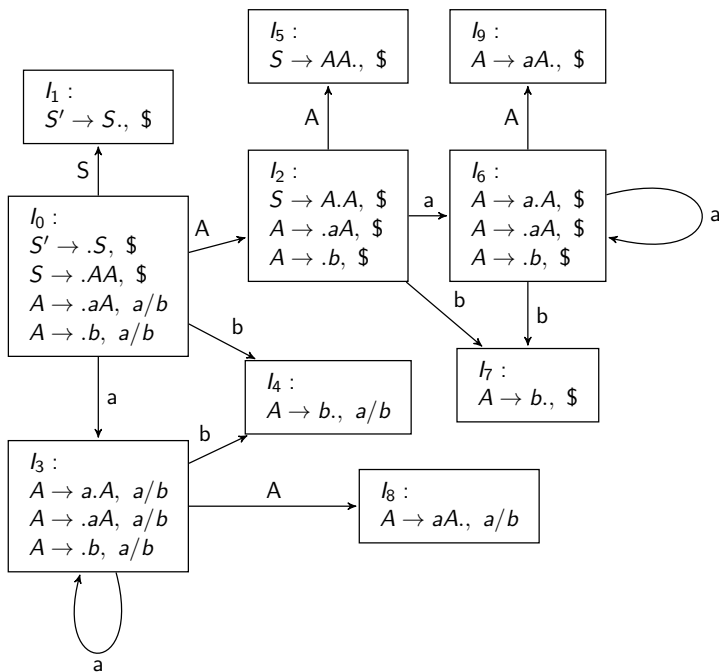
Exercise: Build LR(1) Parser

$$S' \rightarrow S$$

$$S \rightarrow AA$$

$$A \rightarrow aA$$

$$A \rightarrow b$$



Exercise

Parser actions on input

abab\$

bb\$

aabaaaab\$

aab\$

LALR (*lookahead* LR) Parsing

Often used in practice for efficiency reasons

- many fewer states than LR(1) parser (same number of states as SLR)
- combines LR(1) states having set of items with the same first components (*core*)

LALR (*lookahead* LR) Parsing

- Most common syntactic constructs of programming languages can be expressed by LALR grammar.
- There are a few constructs that cannot be handled by SLR techniques.

Example

$$S' \rightarrow S$$

$$S \rightarrow AA$$

$$A \rightarrow aA$$

$$A \rightarrow b$$

- form the set of LR(1) states
- identify pairs of states with the same *core*, but different lookaheads

Example

$$S' \rightarrow S$$

$$S \rightarrow AA$$

$$A \rightarrow aA$$

$$A \rightarrow b$$

LR(1) states:

- three pairs of identical states - states with the same core, but different lookaheads
- merge together identical states to get a parser with lesser number of states

Merging States

A pair of states with the same core:

$$\{[A \rightarrow a.A, \$], [A \rightarrow .aA, \$], [A \rightarrow .b, \$]\}$$
$$\{[A \rightarrow a.A, a/b], [A \rightarrow .aA, a/b], [A \rightarrow .b, a/b]\}$$

Merge the two states into a single LALR state (combine the lookaheads):

$$\{[A \rightarrow a.A, a/b/\$], [A \rightarrow .aA, a/b/\$], [A \rightarrow .b, a/b/\$]\}$$

A state transition in LR(1) to any of the original states is made to the merged state in LALR.

Pair of states with same core

$I_3 :$

$A \rightarrow a.A, a/b$

$A \rightarrow .aA, a/b$

$A \rightarrow .b, a/b$

$I_6 :$

$A \rightarrow a.A, \$$

$A \rightarrow .aA, \$$

$A \rightarrow .b, \$$

Pair of states with same core

$I_8 :$
 $A \rightarrow aA., a/b$

$I_9 :$
 $A \rightarrow aA., \$$

Pair of states with same core

$l_4 :$
 $A \rightarrow b., a/b$

$l_7 :$
 $A \rightarrow b., \$$

Combining Pair of states with same core

$I_3 :$

$A \rightarrow a.A, a/b$

$A \rightarrow .aA, a/b$

$A \rightarrow .b, a/b$

$I_6 :$

$A \rightarrow a.A, \$$

$A \rightarrow .aA, \$$

$A \rightarrow .b, \$$

$I_{36} :$

$A \rightarrow a.A, a/b/\$$

$A \rightarrow .aA, a/b/\$$

$A \rightarrow .b, a/b/\$$

Combining Pair of states with same core

$I_8 :$

$A \rightarrow aA., a/b$

$I_9 :$

$A \rightarrow aA., \$$

$I_{89} :$

$A \rightarrow aA., a/b/\$$

Combining Pair of states with same core

$I_4 :$

$A \rightarrow b., a/b$

$I_7 :$

$A \rightarrow b., \$$

$I_{47} :$

$A \rightarrow b., a/b/\$$

Exercise

Build the LALR parsing table for the example grammar and see the action of the parser on inputs *aabb*\$ and *aa*\$.

Conflicts due to merging

Given an LR(1) grammar (no parsing-action conflicts).

- no shift/reduce conflict in LALR
- merger may result in a reduce/reduce conflict (see example 4.58 in ALSU)

LALR

- correct input - behaves same as LR(1)
- erroneous input - LALR may proceed to do some reductions after LR(1) has detected the error, but LALR will eventually detect the error.

Exercise

Check if the following grammar is LALR? If YES, build LALR parsing table for the grammar.

$$S \rightarrow L = R \mid R$$

$$L \rightarrow *R \mid id$$

$$R \rightarrow L$$

References:

- Aho A.V., Lam M.S., Sethi R., and Ullman J.D. Compilers: Principles, Techniques, and Tools (ALSU). Pearson Education, 2007.

Further reading:

- ALSU chapter 4, section 4.7